

Optical access networks to support future 5G and 6G mobile networks [Invited]

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This article presents an extended version of the ECOC 2024 invited paper on future converged fixed/mobile access networks in the 6G era. First, we outline the expectations for 6G from the perspective of a European operator, and then explore the potential optical network solutions that will support the future evolution of mobile networks. We identify key factors such as enhanced connectivity, efficiency, and flexibility, which will be enabled by technologies including AI, edge computing, HS-PON/PtP, VHS-PON/PtP, all-photonic networks, extended CTI, Open RAN, and other emerging features. © 2025 Optica Publishing Group. All rights, including for text and data mining (TDM), Artificial Intelligence (AI) training, and similar technologies, are reserved.

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1. INTRODUCTION—6G IN A NUTSHELL

The sixth generation (6G) of mobile networks is being developed with a strong focus on sustainability and inclusivity, aiming to balance the challenges of reducing the environmental impact while extending connectivity to underserved populations. It promises to deliver ubiquitous intelligence, enhanced security, and resilience. Building on the advancements of 5G, six usage scenarios have been identified for 6G, as depicted in Fig. 1(a): enhanced mobile broadband (eMBB) will evolve toward immersive communication; massive machine-type communication (mMTC) will transition into massive communication; ultra-reliable and low-latency communication (URLLC) will advance to hyper-reliable and low-latency communication (HURLLC); ubiquitous connectivity, artificial intelligence (AI) and communication, and integrated sensing and communication will introduce three novel application domains for 6G [1–3].

Figure 1, produced by the International Telecommunication Union Radiocommunication Sector (ITU-R) for International Mobile Telecommunications-2030 (IMT-2030) [1], illustrates the “palette diagram” of 6G capabilities, outlining research targets for exploration within the IMT-2030 framework. It outlines 15 key capabilities to guide future research and development. These include enhancements to 5G features, such as improved peak data rates (up to 1 Tbps), user-experienced data rates (1–10 Gbps), spectrum efficiency (unspecified improvement), connection density (10^6 – 10^8 devices per km^2), mobility (500–1000 km/h), latency (0.1–1 ms), reliability (99.999%–99.99999%), and security. New targets include expanded coverage, sensing integration, AI optimization, sustainability, interoperability,

and precise positioning (1–10 cm). These capabilities aim to meet the demands of advanced communication systems while addressing environmental and technological challenges. The palette diagram visually represents these goals, setting benchmarks for 6G’s development.

To be more specific in terms of possible use cases, 6G could enable immersive applications such as holographic communications, ultra-precise digital twins, and advanced AI-driven services, which demand ultra-low latency, high reliability, and centimeter-level positioning. These use cases must demonstrate clear benefits in sectors such as healthcare, smart cities, education, and entertainment, ensuring that the transition to 6G brings measurable improvements to daily life and industry.

Moreover, mobile and fixed network operators emphasize the importance of ensuring that 6G enables a seamless migration from 5G while maintaining compatibility with earlier generations. Orange, an incumbent operator for both fixed and mobile networks, has presented initial insights into what 6G could represent for operators [4]. This document has been published after the final writing of the ECOC2024 invited paper [5], which is extended here. In this white paper, Orange highlights the need for a smooth, evolutionary progression in mobile network technology beyond 2030, avoiding major hardware overhauls. The introduction of 5G has not significantly accelerated device renewal within Orange’s footprint. Moreover, the technical requirements of the most widely used services on mobile networks, such as social networks and video streaming, remain largely compatible with 4G data rates. Many benefits of 5G, including low latency and high capacity, have yet to deliver visible advantages for end users. For industries and verticals, however, the untapped potential of 5G presents

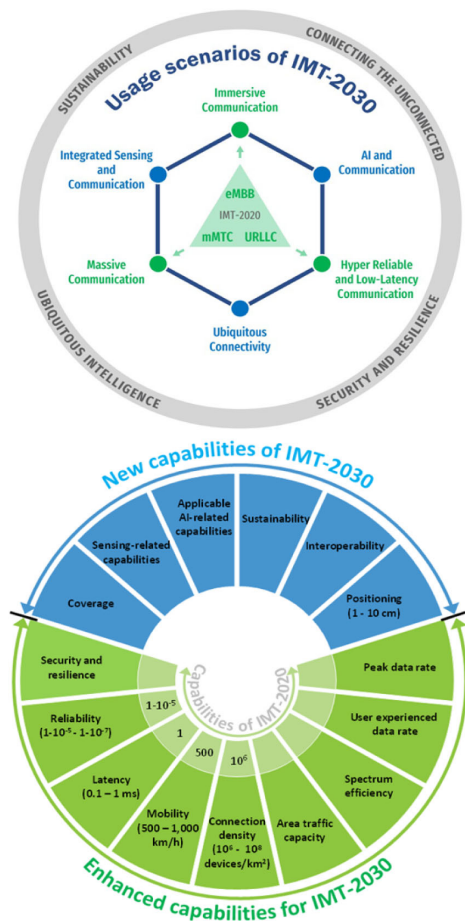


Fig. 1. 6G definition by ITU-R as IMT2030 with six usage scenarios (three new) and four overarching aspects of 6G (top); palette diagram of capabilities including six new ones (bottom). Source ITU-R 2023 [1].

both significant opportunities, such as enhanced automation and IoT applications, and risks, including challenges for traditional operators competing with private networks. Furthermore, delays in stand-alone 5G deployments have slowed the realization of 5G's full capabilities, which highlights the need for operator-led strategies to overcome these barriers. As a result, future network evolutions must focus on delivering tangible value while prioritizing sustainability.

While 5G stand-alone networks address many current needs, Orange identifies six key areas where 6G could provide additional value beyond 5G: improved productivity, reduced CO₂ emissions, sustainable energy production, more efficient healthcare systems, better agricultural practices, and expanded access to knowledge and culture. These advances stem from 6G's anticipated innovations, such as native AI integration, advanced sensing capabilities, and greater energy efficiency, which aim to address challenges that may surpass the capabilities of stand-alone 5G. Sustainability is described as a balance of three critical societal dimensions: environmental, to preserve ecosystems for humans and other living species; economical, to ensure sufficient revenue for all ecosystem players; and social, to foster trustworthiness, inclusivity, and relevance in society. Importantly, new network features may be reconsidered or avoided altogether if their deployment jeopardizes net

zero carbon targets, which many industries aim to meet by 2040 [6].

In parallel, Orange continues to invest in 5G radio access networks (RANs) with deployment plans extending to 2030. These deployments include the installation of the so-called New Radio using carriers operating at 3.5 and 2.1 GHz, with or without RAN sharing [7]. Moreover, recent studies indicate that mobile traffic growth is slowing globally [8]. It shows that global mobile traffic increased by only 21% year-over-year in the third quarter, a slowdown compared to 25% last year, 40% three years ago, and a striking 95% six years ago. Ericsson predicts a gradual decline in growth to 16% by 2030. However, these figures also account for fixed wireless access (FWA), which reflects notably different user behavior. From an operator's perspective, the slowdown in mobile traffic growth stems from market saturation in device adoption, slower uptake of 5G-specific services, and more efficient data usage through advancements in compression and streaming technologies. Economic constraints and a shift to Wi-Fi for bandwidth-heavy applications further limit the growth, while stabilized usage patterns and slower device replacement cycles reduce the demand for new data-intensive applications. These factors collectively temper the pace of traffic increases despite ongoing network advancements. Nevertheless, global traffic continues to grow.

As a result, the next generation of mobile networks must deliver incremental improvements in capacity and performance without excessive increases. Table 1 provides a tentative proposal for key performance indicators (KPIs) for 6G extracted from [4]. These indicators highlight the anticipated enhancements in 6G compared to 5G, such as quite similar data rates and latency experienced by users but improved the area capacity, connection density, positioning accuracy, as well as greater energy efficiency. This should enable hyper-reliable communication and new use cases for 6G to support diverse environments such as dense urban areas, wide outdoor spaces, and high-speed mobility.

Achieving this will rely on technological enablers to enhance capacity per cell. Innovations in multiple-input multiple-output (MIMO) technology, such as higher-order configurations and narrower beamforming, will support multi-user scenarios within each cell.

In the evolution of 5G, the exploitation of frequency bands FR1 (0.41–7.125 GHz) and FR2 (24.25–71 GHz) has been instrumental. Looking ahead to 6G, discussions at ITU meetings had highlighted the potential of a third frequency range, FR3 (7.125–24.25 GHz), also known as the upper mid-band. This new spectrum offers significant opportunities for mobile communication technologies. Enhanced MIMO and expanded frequency ranges will enable 6G to achieve a tenfold increase in data rate capacity. More recently, in the first discussions toward the first specifications in 3GPP in 2030, the 6 GHz range is seen as the prime band that should be secured with an exclusive licensed mode, and the 7–15 GHz range would be a potential complement. This should allow native spectrum sharing between 5G and 6G, i.e., without minimal overheads.

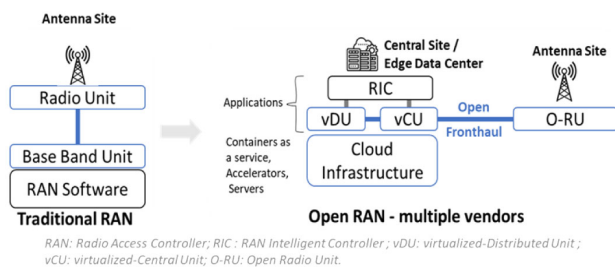
To improve radio quality, adapt to diverse scenarios, and collect environmental data, 6G networks will integrate advanced

Table 1. Tentative KPIs for 6G

KPI	Possible Extreme Value	5G Reference [9]	Complement, e.g., Target Scenario
User experienced data rate ^a (at cell stage)	300 Mbit/s	300 Mbit/s	Dense urban
	100 Mbit/s	100 Mbit/s	Other outdoor environments
Area capacity ^b	3 Tb/s/km ²	750 Gb/s/km ²	Dense urban
	450 Gb/s/km ²	100 Gb/s/km ²	Outdoor and wide area
Connection density	35, 000/km ²	25, 000/km ²	Mobile broadband-dense urban
	15, 000/km ²	10, 000/km ²	Mobile broadband-urban macro
	1.10 ⁶ /km ²	1.10 ⁶ /km ²	Massive Internet of Things (IoT)
Positioning accuracy	<10 cm	1 m	Indoor deployment
	<1 m	3 m	Outdoor and wide area
Energy efficiency	×10 versus 5G	No quantitative requirement	At least as much as capacity increase so network energy consumption remains stable or decreases
Minimum end-to-end latency	5 ms	N/A	In generic deployments for services requiring it
	0.5 ms (URLLC)	(Not applicable)	For specific services and use cases associated with specific deployments
Reliability	99.9%	N/A	For most services, typically mobile broadband
	99.999%	Idem	For specific services and use cases associated with specific deployments
Mobility	500 km/h	Idem	For specific services (e.g., very high-speed trains and planes)

^aNote: 250 Mbit/s required for immersive experiences. Most future usages need <100 Mbit/s.

^bNote: 30% activity factor assumed.

**Fig. 2.** Evolution from traditional RAN to Open RAN.

sensing tools. Sub-THz frequency bands could be of particular interest, offering wide coverage and the ability to penetrate obstacles like walls. Additionally, satellite communication may become part of the 6G radio access network (RAN), ensuring seamless real-time connectivity and extending coverage to unconnected regions. High-speed public radio access points at locations such as transportation hubs or shopping centers could support mmWave small cell backhaul. However, ensuring compatibility with cellular services and addressing security concerns remain critical challenges for 6G development. The technology will benefit significantly from the virtualization of core network functions (entirely) and RAN functions (partially), building on the current trends of increasing network softwarization and Open RAN initiatives for enhanced security, interop, latency, and services, as illustrated in Fig. 2 and discussed in [10].

Designed as the mobile network of the 2030s, cloud-native and artificial intelligence (AI)-native architectures toward a level 4 automation will be a key enabler of the future 6G. While these architectures have immense potential to optimize network efficiency and performance, it is critical to address their energy demands. The development of sustainable AI models, energy-efficient data centers, and innovative green

technologies will be essential to align extensive AI use with 6G's sustainability goals. It will integrate advanced AI and computing capabilities into both the network and its services. Also, to maximize efficiency and flexibility, 6G hardware and software features should be managed within cloud instances wherever possible.

From a broader perspective, the ICT sector accounts for up to 4% of global carbon emissions, according to the ITU [11], and the growing use of AI is likely to increase this share in the future. Orange, among other major industrial players in the telecommunications sector [6], has committed to achieving net zero carbon emissions by 2040. This goal requires targeted actions to reduce carbon emissions across Scope 1, Scope 2, and Scope 3, as defined by the Greenhouse Gas (GHG) Protocol established by the United Nations.

In 2023, 92% of Orange's carbon emissions were attributed to Scope 3, highlighting the significant role of indirect emissions in its overall footprint [12]. Figure 3 (left) provides a breakdown of the carbon emissions of Orange France in 2023. From the perspective of an operator, networks accounted for 50%–60% of its carbon footprint, largely driven by energy consumption during the manufacturing and installation of mobile networks [13]. Figure 3 (right) focuses on the energy consumption of Orange France in 2023 by sector of activity, further emphasizing the substantial impact of networks. This energy consumption is directly linked to Scope 2 and Scope 3 of carbon emissions, reinforcing the critical importance of addressing network-related activities to reduce the overall environmental impact.

Therefore, a key focus in the 6G design remains energy efficiency, which has consistently been a primary objective for both network infrastructure and user devices. Beyond energy efficiency, it is essential to minimize total energy consumption,

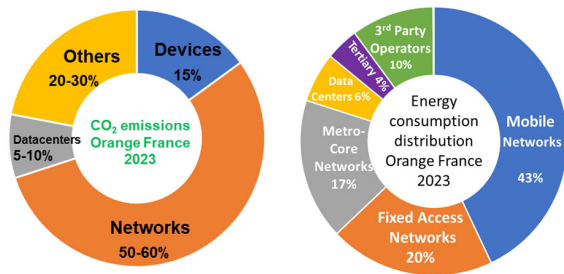


Fig. 3. Orange France distribution of CO₂ emissions [12] (left) and energy consumption per sector of activity in 2023 (right).

reduce greenhouse gas emissions, and promote a circular economy for sustainable development. Achieving this will require the creation of power-efficient and zero-emission 6G technologies for both backhaul and local access networks, with support for small-scale renewable energy sources. The following section describes the possible evolution of fixed access networks to support future mobile technologies.

2. FIXED ACCESS TECHNOLOGIES FOR 5G AND 6G MOBILE X-HAUL NETWORKS

Since the deployment of 5G, fiber has become the preferred solution for connecting antenna sites to fixed networks. This is typically achieved through Ethernet switches equipped with typically 10G bit/s transceiver, connecting to aggregation or metropolitan networks. Instead of being linked directly to an Ethernet switch, mobile antennas can be linked to centralized aggregation equipment, such as optical line terminals (OLTs) already in use for FTTH. The fiber segment connecting the antenna is referred to as a mobile X-haul segment, where “X” can represent backhaul, mid-haul, or fronthaul, as detailed in [14].

The specific X-haul segment depends on service demand characteristics, functional splits of RAN equipment, and constraints at either the antenna or central site. For 6G, new functional splits should be discussed in 3GPP, possibly without great interest in the centralized unit (CU)–distributed unit (DU) split. Then, future optical access networks will need to meet key requirements, including support for ultra-high transmission capacity (potentially in the terabit-per-second range), deterministic low latency, minimal jitter (comparable to 5G), and cost- and energy-efficient X-haul solutions.

Regarding 2G, 3G, 4G, and 5G, most antenna sites are connected via dedicated point-to-point (PtP) fibers as mobile backhaul. With 6G, this configuration is expected to remain largely unchanged, except in specific cases such as indoor coverage for campuses, shopping malls, and hotspots, where a specialized optical backbone may be required for mobile X-haul.

OLTs are key equipment to support future mobile networks evolution as fiber-to-the antenna (FTTA), along with ensuring fiber-to-the-home (FTTH), enterprise (FTTE), or office (FTTO). From this equipment at the central office, point-to-point (PtP) wavelength division multiplexing (WDM) and time division multiplexing/multiple access passive optical

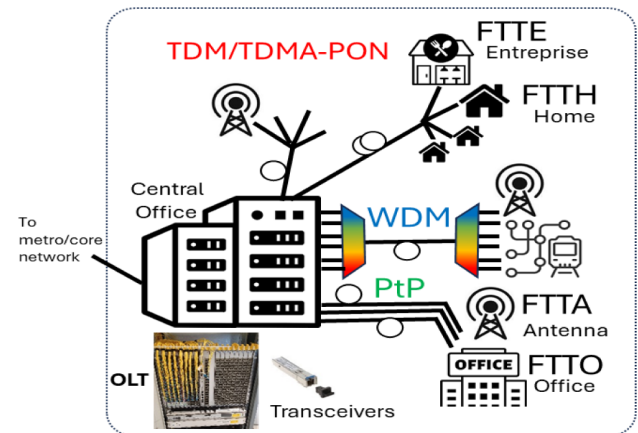


Fig. 4. Optical access networks for mobile X-haul.

network (TDM/TDMA-PON) technologies could support mobile X-haul network evolutions, as depicted in Fig. 4.

For FTTA and FTTO, a PtP “logical” (PtP or WDM) connectivity is preferred, as it provides the full bandwidth of each PtP link to termination. The latency is primarily limited to signal propagation in the fiber (5 μ s/km), with negligible additional latency introduced by signal processing (e.g., FEC and DSP) required to enhance transmission for high bitrates or long reach.

PtP links are evolving toward bidirectional transmission up to 100 Gbit/s over 10, 20, or 40 km of fiber reach [14,15]. Several initiatives in standard bodies are encouraging the development of such solutions, such as ITU-T G.9806 (higher speed bidi PtP up to 100G), IEEE 802.3 cp (50GBASE-BR10/20/40), IEEE 802.3dk (100GBASE-BR up to 40 km, currently ongoing), and 100G Lambda Multi-Source Agreement (100G and 400G PAM4 based on 100G per channel for short reach). Discussions on 200 and 400 Gbit/s are also underway in these standard groups. With such capacity and commercially available high-speed interfaces, PtP links are anticipated to remain the preferred choice for extensive local connectivity to mobile networks, while WDM-PON or TDM-PONs could be alternatives in areas with either limited fiber availability or extended reach fiber network.

When fiber links become scarce, X-haul based on WDM would become interesting. WDM solutions for mobile networks have emerged at up to 25 Gbit/s [16], with commercial solutions available in the C-band and now with ITU-T standard specifications also in the O-band. These include G.698.5, which uses LAN WDM-like spacing (800 GHz), and G.698.6, which specifies medium WDM (7 nm) or coarse WDM (13 nm) spacing. For higher bitrates, 50 or 100 Gbit/s, transmissions in the O-band would become imperative to limit chromatic dispersion penalties. Otherwise, to maintain dense WDM options in the C-band, digital signal processing or more complex optics and coherent transmissions could be required. In any case, the use of PAM4 is a likely first option to consider, as it is well adopted for PtP interfaces already. As of today, WDM transceiver technology beyond 25 Gbit/s per channel is still under development or study.

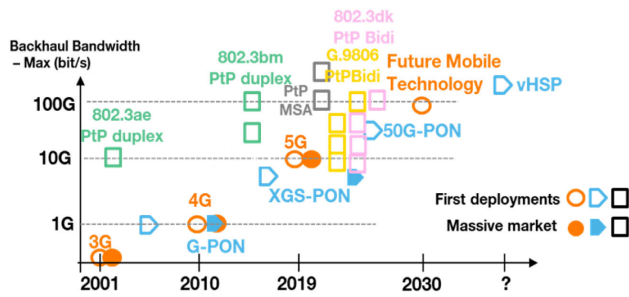


Fig. 5. Optical access networks for mobile X-haul.

For FTTE (small enterprises such as restaurants, grocery shops, etc.) and FTTH (residential users), the high number of terminations requiring fiber connectivity makes time TDM/TDMA PONs a favorable choice. In these systems, the maximum throughput is shared among multiple users, with dynamic bandwidth allocation (DBA) implemented in the upstream to optimize traffic distribution. The latency in such PON systems depends on factors such as vendor implementation and traffic configuration (e.g., DBA, delay tolerance, guard time, fiber length, and processing) and typically ranges from several tens of milliseconds at a minimum. While 50 gigabit passive optical networks (50G-PONs) are reaching finalization by ITU-T as a future generation of PON deployed in FTTH networks [17], challenges such as cost, power consumption, and complexity persist for higher bitrates [18,19]. Discussions on a 200G-PON technology within very high-speed PON (VHSP) scope suggest support for network evolution beyond 2030. Such high bitrate PtMP-PON could be suitable for mobile backhaul with fiber sharing but they face limitations due to packet delay variation, latency, and strict congestion traffic policy for mobile fronthaul. Specific “private” networks, such as campus and industry, and distributed antenna systems, could offer an opportunity for PON to serve 6G X-haul.

Another critical consideration for the 6G mobile X-haul is the maturity of the optical transport technology. This includes its ability to rapidly achieve mass-market adoption, potentially benefiting from cost reductions enabled by shared markets with high-volume connectivity technologies, such as FTTH or data center networks. Figure 5 illustrates the chronological emergence of fixed and mobile network technologies, distinguishing between initial deployments and large-scale market adoption. This perspective highlights that the only fixed network technology with prior mass-market maturity capable of supporting mobile deployments is PtP connectivity. Technologies such as XGS-PON and 50G-PON became available too late to support initial 5G deployments or the late-stage evolutions of 5G Advanced. Similarly, VHSP is likely to mature later than the anticipated early deployments of 6G. Consequently, we expect that PtP interfaces operating at 800 Gbit/s or beyond, currently maturing for data center applications [20], will play a key role in supporting the future 6G mobile X-haul.

Other initiatives in standard bodies and global forum may also encourage and accelerate the development of optical technologies for 6G X-haul. The fifth generation of fixed networks (F5G) is advancing toward F5G Advanced specifications, aiming to deliver over 10 times higher bandwidth, denser fiber

connections, improved reliability, enhanced energy efficiency, and advanced sensing capabilities with 1 m accuracy and sub 1 ms end-to-end latency, alongside level 4 autonomous networking [21]. F5G Advanced introduces key features such as higher network speeds powered by technologies such as >10 G-fiber-to-the-room and 50G-PON, and the integration of cloud computing into optical networks for on-demand services. It also emphasizes autonomous networking by defining functions, architectures, and interfaces to improve the quality of experience (QoE) through automation and energy-efficient designs. Furthermore, it highlights the potential of optical communication in industrial sectors and the importance of collaboration between networking and computing to optimize service quality and enable low-latency services. With these advancements, F5G Advanced is well-positioned to support future mobile networks, including 5G and 6G. Looking ahead, projections outlined in [16] explore the potential sixth generation of fixed networks (F6G) beyond 2030. The IOWN Global Forum envisions enhancing mobile networks by 2030 through a transition to all-optical connections via all-photonic networks (APNs), extending connectivity from site-to-site to memory-to-memory integration. Metrics proposed in [22] suggest significant improvements, including bandwidth per endpoint exceeding 10 Tbit/s, minimized latency, automated system failure handling, and reduced jitter for superior performance.

To address latency challenges, current cooperative transport interface (CTI) specifications are insufficient for short transmission time intervals (TTIs) and mid-haul/backhaul in upcoming mobile networks. Enhancements to CTI specifications are necessary. IOWN proposes achieving end-to-end low latency through an “extended CTI,” enabling APN controllers to receive mobile data directly from the CU/DU [23]. Additionally, optical switching schemes for dynamic load balancing and power consumption reduction are proposed, integrating APN with RAN intelligent controllers for improved efficiency.

As discussed in the previous section, most of the energy consumption and carbon footprint in mobile networks is primarily associated with the radio segment. However, efforts in optimizing fixed access networks can also contribute to enhancing the sustainability of future 6G systems. Increasing the split ratio of PONs [24] and implementing highly dynamic sleep modes for every network termination, bringing flexibility in point-to-point links [25] and intelligent and coordinated controllers [26] and interfaces, such as the aforementioned, could significantly help in achieving sustainable fixed networks for the future mobile generation.

3. TOWARD FLEXIBLE AND AUTONOMOUS NETWORKS FOR FIXED MOBILE CONVERGENCE

Figure 6 presents a possible example of an optical access network to support future evolutions of mobile networks. Future fixed networks will need to increase their resilience by incorporating greater redundancy to minimize network downtime. Advanced network monitoring capabilities, as well as technologies such as fiber sensing and AI/machine learning,

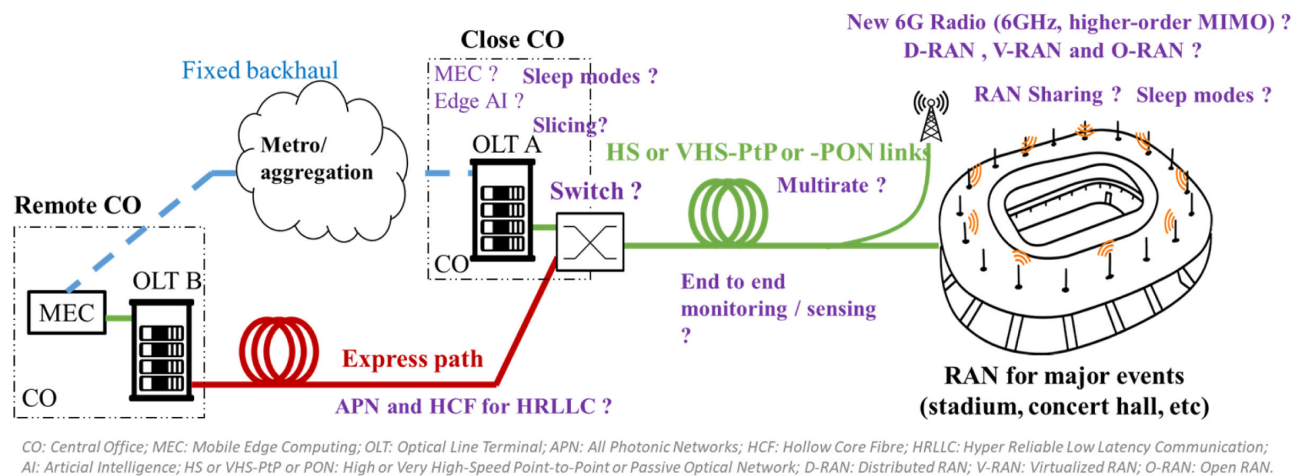


Fig. 6. Schematic view of what could be future access networks for mobile X-haul.

are expected to facilitate the transition toward autonomous networks. The interaction between AI and 6G can be seen in two ways: AI enhancing communication and communication empowering AI. These dual dynamics will open new opportunities for the advancement of 6G networks. Indeed, AI is intended to play a central role in achieving flexible and autonomous networks, particularly in areas such as dynamic resource allocation, anomaly detection, and predictive maintenance. Sensing data, such as user mobility patterns, physical strains on fibers, network traffic loads, and environmental factors, will be collected and analyzed to enable real-time decision-making, network optimization, and maintenance. For example, AI algorithms could predict traffic congestion and preemptively reconfigure the network, ensuring seamless connectivity. Additionally, AI will augment SDN by enhancing its ability to dynamically adapt to network demands, providing the agility needed for fixed-mobile convergence.

In the meantime, distributed RAN, virtualized RAN, and centralization induce major evolutions in the RAN architecture, both on the hardware and software parts [10]. Open RAN (O-RAN) and Cloud RAN can also leverage energy efficiency improvements with RAN architecture evolution. That will certainly impact fixed network evolutions as well. The evolution to O-RAN is a key enabler for RAN sharing and network slicing, which relies on partitioning the network infrastructure into virtualized segments to cater to diverse service requirements, such as bandwidth, latency, and security, within a single physical network that is fully interoperable. This enables efficient resource utilization for any operator and customization of services for various applications, including residential, business, and mobile connectivity, while maintaining isolation and performance guarantees for each slice. As an example of the combined benefits of AI and O-RAN, in [27], the authors introduce an AI-driven framework, designed to facilitate autonomous fault management across multi-vendor and multi-domain environments within O-RAN.

Moreover, the transition from traditional OLT management to software-defined-network-based controllers involves separating management functions from hardware and implementing them in software controllers, with possible

convergence with the radio intelligent controller (RIC). This shift enhances flexibility and scalability by providing centralized and coordinated control and programmability and will lead to more dynamic, efficient, and converged OLT and RAN management.

With more and more network function virtualization, automation, and flexibility expected in future networks, fixed and mobile network convergence will not anymore take place only where physical interconnections exist (today in the central office and at antenna sites). However, this approach may not be optimal in terms of carbon footprint and network sharing, which aim to extend computing and server functions—such as mobile edge computing and edge AI—closer to the user. The optical network's architecture must then mature to support these evolutions toward autonomous networks, with flexibility expected from the physical layer to the protocol layer, to ease transparent end-to-end connectivity.

Enhanced photonic switching with multi-Tbit/s capacities for next generation central offices is being developed in the Octopus project with a dedicated software defined networking (SDN) controller to support time-sensitive networking (TSN) for future mobile networks. An optical bypass of the OLT with PtP, demonstrating flexible architectures for low latency and power savings capabilities, has been demonstrated in [25,26].

Recently, new designs of hollow-core fibers demonstrated attenuation profiles with transmission losses lower than 0.11 dB/km [28]. These innovative fibers could potentially contribute to reduce latency to 3.5 μ s per km of fiber (instead of 5 μ s/km with SSMF) in transport networks for mobile. However, improvements in the production process (longer spools, fiber coupling, soldering processes, and spectral band extension) are still necessary to demonstrate significant interest for the deployment of new fibers.

4. CONCLUSION

Future convergence in fixed and mobile access networks within the 6G era will be present in many ways in the coming decade. This convergence promises enhanced synergy and seamless integration between diverse network infrastructures, enabling

improved connectivity, performance, efficiency, and flexibility. By leveraging advanced technologies such as end-to-end network slicing, edge computing, and AI-driven optimization, 6G networks will foster the necessary evolution of optical transport with converged control, monitoring, and orchestration.

The arrival of 6G in a context of optimizing operators' investments in mobile services positions the optical line terminal (OLT) as a pivotal component for traffic aggregation. The OLT shelf, as the first aggregation point for both fixed and mobile traffic, provides opportunities for innovative network architecture optimization. These scenarios could rely on existing PONs and PtP interfaces, or even explore new all-optical OLT shelf architectures designed to meet the specific needs of emerging 6G services. Such advancements will be critical for supporting a wide range of 6G applications, including ultra-reliable low-latency communications (URLLCs), massive IoT deployments, and immersive experiences.

However, significant challenges must be addressed to fully realize the potential of this convergence. Issues such as standardization, interoperability, and security remain central to the successful deployment and operation of 6G networks. Collaborative efforts among industry stakeholders, research communities, and regulatory bodies will be essential to overcome these obstacles and establish frameworks that ensure the reliability and scalability of converged infrastructures.

In conclusion, the convergence of fixed and mobile networks within the 6G ecosystem represents a transformative step toward a hyper-connected future. By building on the strengths of current technologies and addressing key challenges, this evolution not only enhances the capacity and flexibility of network architectures but also paves the way for groundbreaking applications that will redefine industries and elevate global connectivity. This journey will require sustained innovation, cooperation, and investment to unlock the full potential of the 6G era.

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